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Accuracy of the revised Addenbrooke Cognitive Examination (ACE-R) in older adults with low education and mild cognitive impairment: results of a cross-sectional study in two metropolitan areas of Northeast Brazil

Acurácia do Addenbrooke-versão revisada (ACE-R) em idosos com baixa escolaridade e comprometimento cognitivo leve: resultados de um estudo seccional em duas áreas metropolitanas do Nordeste do Brasil

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ABSTRACT

Objective: To determine the diagnostic accuracy of the Addenbrooke's Cognitive Examination (ACE-R) for older adults with low education, without dementia, in two capitals in northeastern Brazil, compared to subjects with MCI. **Methods:** 100 participants were collected from a previous neurological and psychiatric evaluation and were subsequently subjected to the ACE-R. Among them, 18 subjects with amnesic mild cognitive impairment (aMCI), 22 with non-amnesic mild cognitive impairment (naMCI), and 60 healthy controls. **Results:** Optimal ACE-R accuracy yielded excellent values for the comparison between controls and naMCI [Area Under the Curve (AUC) = 0.919] and controls and aMCI (AUC= 0.921); conversely, very fair accuracy was reported for the comparison between aMCI and naMCI (AUC= 0.578). **Conclusions:** These findings support establishing reliable cutoff scores for cognitive assessment of elderlies with low schooling and cognitive decline, not dementia, taking into consideration ecological and regional variables.

KEYWORDS

Mild cognitive impairment; cognitive screening; illiterate; Addenbrooke, accuracy.

RESUMO

Objetivo: Determinar a acurácia diagnóstica do Exame Cognitivo de Addenbrooke (ACE-R) para idosos com baixa escolaridade, sem demência, em duas capitais no nordeste do Brasil, comparando a sujeitos com CCL. **Métodos:** Foram coletados 100 participantes a partir de uma avaliação neurológica e psiquiátrica prévia, sendo submetidos aos ACE-R posteriormente. Dentre eles, 18 sujeitos com comprometimento cognitivo leve amnésico (aCCL), 22 com comprometimento cognitivo leve não amnésico (naCCL) e 60 controles saudáveis. **Resultados:** Os pontos de acurácia do ACE-R foram considerados excelentes para a comparação entre controles e naCCL [Área sob a curva (AUC) = 0,919] e controles e aCCL (AUC= 0,921); por outro lado, foi relatada uma baixa acurácia para a comparação entre aCCL e naCCL (AUC= 0,578). **Conclusões:** Os achados dão suporte à necessidade de estudos estabelecendo pontos de corte confiáveis para a avaliação cognitiva de idosos com baixa escolaridade e declínio cognitivo sem demência, levando-se em consideração variáveis ecológicas e regionais.

PALAVRAS-CHAVE

Comprometimento cognitivo leve; triagem cognitiva; analfabetos; Addenbrooke, acurácia.

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INTRODUCTION

The increase in life expectancy in Brazil has been associated with a higher prevalence of age-related mental conditions such as Alzheimer's Disease (AD)^{1,2}. According to the World Health Organization of the United Nations (WHO-UN), Brazil is one of 10 countries with the world's largest population of older adults. Data from the Brazilian Institute of Geography and Statistics (IBGE) show that in Brazil, older adults aged 65 or older represented 10.9% of the total population in 2022³, an increase of 57.4% to the former Census (2010)⁴, with impact on the population aging index, which in 2022 reached 55.2³.

According to WHO data⁵, in 2022, around 55.2 million people were living with dementia in the world; populational projections in the following years expect a significant prevalence increase and report there will be approximately 153 million cases by 2050⁶. Moreover, it is estimated that 4.5 million people in Latin America (LA) were living with dementia in 2019⁷, and more than 40% of them were Brazilians⁸; in the Northeast city of Fortaleza, it is estimated that at least 23,000 adults have dementia⁴. In Brazil, among the most relevant risk factors for cognitive disorders, low education stands as the most significant population-attributable fraction, around 7.7%⁸.

Mild cognitive impairment (MCI) is deemed an intermediate stage between healthy aging and pathological aging^{9–12}. The prevalence of MCI in people over 65 years of age is 12–18%¹³; around 50% of MCI subjects may develop dementia within three years, and from baseline (MCI diagnosis), the annual conversion to dementia (all types) ranges from 6–15%^{14–16}. In Brazil, the incidence rate of MCI is 13.2%¹⁷. Clinical subtypes of MCI comprise amnesic, non-amnesic, and multi-domain impairment (aMCI, naMCI, and mdMCI, respectively). The definition of aMCI by Petersen encompasses a presumed degenerative etiology, likely representing a preclinical form of Alzheimer's disease⁹. The annual rate of progression from aMCI to AD is estimated at 10–15%^{10,18}. Conversely, naMCI may include heterogeneous conditions, with less characteristic neuroimaging and neuropsychological profile²⁰ and a wide range of neuropathological findings²¹, including vascular disease, lobar frontal degeneration, and Parkinson-related syndromes^{9,19}.

Cognitive tests are commonly used to screen cognitive impairment, diagnose etiologically, establish disease severity, and monitor disease progression²². A significant challenge for the initial assessment of age-related cognitive disorders is to select a screening test that is both sensitive and specific for differential diagnosis. Ceiling and floor effects limit the ability of a test or some of its items to assess cognitive impairment accurately²³. The ceiling effect occurs when score distribution

is skewed, and variance in a cognitive domain is no longer “achieved,” thereby preventing assessment test performance. This effect has been reported in several studies and is primarily related to educational background.

Few cognitive screening studies outside wealthier metropolitan areas in Brazil, including most northeastern cities, have been conducted with non-dementia older adults, whether using MMSE or ACE-R³². The relative scarcity is mainly because these studies involve expensive assessments of specialized services. We conducted an electronic search in the PUBMED database; we found only two studies conducted in Northeast Brazil—one from 2005 by Brito-Marques and Cabral Filho³³ and another one from 2012 by Caldas et al.³⁴. Cognitive assessment of adults who are either illiterate or with low levels of education poses additional challenges. About 14 million people, most of them older adults (20.4%), are estimated to be illiterate in Brazil, and there is significant controversy about reliable approaches for age-related cognitive assessment of this population. Previous studies have sought to establish valid cutoff scores for illiterate adults^{1,2}, but there is no consensus on whether data can be replicable in populations from different regions nationwide. Sociocultural aspects, e.g., living in poor resource areas and having limited access to the Internet, banking services, and public transportation, may influence cognitive performance³⁵. For most old-age public services in Brazil, complete neuropsychological batteries are unavailable. Therefore, reliable, rapid, and easy-to-use tools are required to provide an accurate screening of MCI subtypes.

Among the standard instruments to screen for cognitive decline, the Mental State Exam (MMSE) and the Addenbrooke's Cognitive Examination-Revised (ACE-R) are both relevant in general practice and specialized services and officially recommended by the Brazilian Academy of Neurology²⁴. A superior diagnostic accuracy of the ACE-R compared to the MMSE, with less educational bias and broader cognitive evaluation, has been suggested²⁵. ACE-R was validated in 2012 by Amaral-Carvalho and colleagues and contains suggestions for data interpretation according to literacy level¹⁹. In a study by Almeida²⁸, MMSE cutoff scores of 23–24 yielded 84% sensitivity (sn) but low specificity (sp) (60%) to discriminate between MCI-normal aging and dementia. Brucki²³ reported that educational level was the most critical factor influencing MMSE scores (ANOVA $F [4, 425] 100.45, p < 0.0001$), and cutoff scores for illiterate people (score = 20) were the lowest across all groups evaluated. A previous finding of our group identified lower scores for healthy controls in ACE-R compared to other Brazilian studies [for a thorough review, see Tavares-Júnior et al. (22)]. Evidence has also indicated ACE-R as a valuable tool for differentiating between healthy controls and MCI^{30,31} and predicting conversion to dementia from amnesic MCI (aMCI)³¹.

The study of psychometric properties of ACE-R and other cognitive instruments outside wealthier metropolitan areas may contribute to the early detection of aMCI and naMCI, making their use practical and adapted to the local reality³⁴, particularly in Northeast Brazil, as well as to avoid educational bias usually seen in cognitive scales validated to the Brazilian population. Our study aimed, thus, to determine the diagnostic accuracy of the ACE-R as a cognitive screening tool for older adults with low levels of education and healthy aging and subtypes of MCI in Brazil. This study will also be supported by data from our research project—the Addencog project²⁹, a multicenter initiative in two metropolitan areas in Northeast Brazil. Our central hypothesis, also based on our previous findings, is that the ACE-R test has good accuracy in MCI subtypes, and the specific cutoff points of this test can be used for cognitive assessment of older people with low levels of education residing in Northeast Brazil.

METHODS

Participants

Participants of the Addencog research project were older adults (over 60 years) from two Brazilian municipalities, consecutively recruited in 3 community centers from January 2018 to October 2021, as well as patients attending the geriatric neuropsychiatry outpatient services of the Federal University Hospital of Ceará (Fortaleza, state of Ceará) and the Nina Rodrigues Hospital (São Luís, state of Maranhão). The present study evaluated a total of 100 participants comprising 22 with naMCI (08 from Fortaleza and 14 from São Luís), 18 with aMCI (4 from Fortaleza and 14 from São

Luís) and 60 healthy aging controls (14 from Fortaleza and 46 from São Luís). Sociodemographic information was collected from medical history and clinical examination through an interview conducted by a geriatric psychiatrist and three senior neurologists (GSA, JISN, PB, and WL). The clinical exam was performed for all subjects and included a psychiatric interview based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), a neurological exam, and a laboratory investigation when required (Figure 1)³⁶.

Experimental Procedures

All participants underwent neurological and psychiatric examination and imaging studies (computerized tomography or magnetic resonance imaging). All subjects, patients, and their family members were evaluated using the Clinical Dementia Rating (CDR), MMSE, and the Brazilian-validated version of the Pfeffer Activities Questionnaire^{37,38} to assess functional ability: the CDR assesses cognition and behavior in a structured questionnaire with closed questions that verify subject performance of daily living activities. It examines six categories: memory, orientation, problem-solving, social relationships, leisure activities or the family environment, and personal care. The Pfeffer Activity Questionnaire, consisting of 11 questions, is an instrument applied to the caregiver or close family member of the older adult to assess performance in functional daily activities. Patients were categorized as cognitively healthy and MCI if CDR = 0 and CDR = 0.5, respectively. All patients in the study exhibited scores of < 6 on the Pfeffer Questionnaire.

MCI diagnosis was based on Petersen criteria⁹. Healthy and MCI participants showed Pfeffer scores lower than 5,

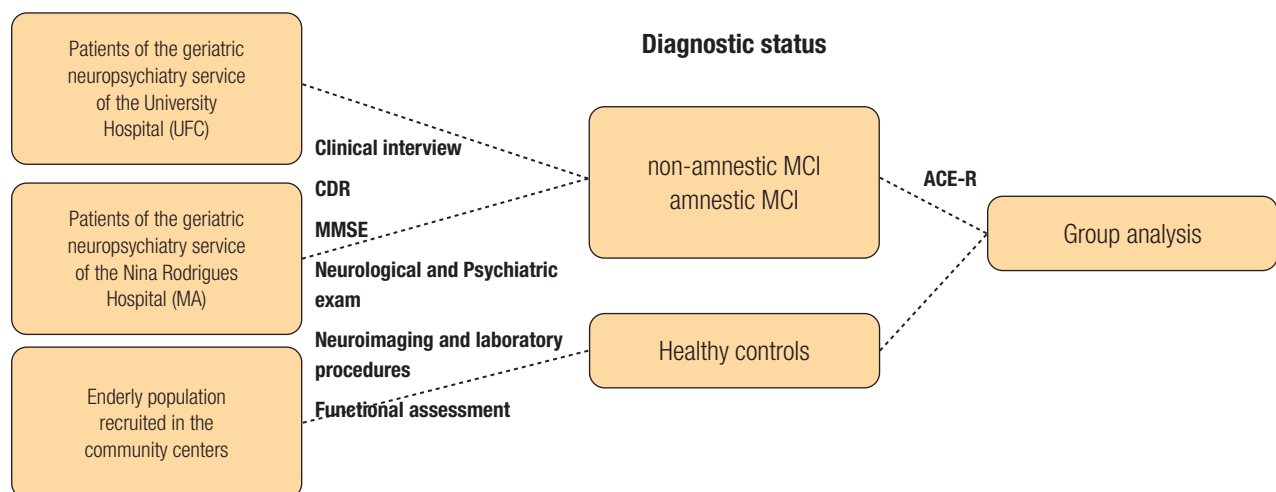


Figure 1. Flowchart depicting the study design

Adapted from Tavares-Júnior et al., 2019

Abbreviations: CDR – Clinical Dementia Rating scale; MCI – Mild Cognitive Impairment; UFC – Universidade Federal do Ceará, MMSE – Mini-Mental State Exam; ACE-R – Addenbrooke's Cognitive Examination Revised.

which indicates normal functional ability. To determine MCI status and subtypes, we adapted from Brucki and Nitrini³⁹ weighted scores (z scores) of MMSE subsets; patients were labeled aMCI if memory-related domains (immediate memory-recall) were lower than 1.5 SD and non-amnesic MCI (naMCI) if any other domain(s) rather than memory was (were) lower than 1.5 SD⁹. The ACE-R was administered to all participants and assessed different cognitive domains, including memory, attention, language, verbal fluency, and visuospatial skills⁴⁰. Only subjects with up to 5 years of formal education, verified by family members, were included in the study. To exclude the diagnosis of dementia, DSM-5, NINCDS-ADRDA⁴¹, and International Statistical Classification of Diseases and Related Health Problems (ICD-10) criteria were employed^{36,42}. The main exclusion criteria were dementia, history of stroke, traumatic brain injury, epilepsy, multiple sclerosis, or previous psychiatric conditions, such as major depression, bipolar disorder, schizophrenia, or alcohol dependence.

Ethical statement

The study was approved by the National Research Ethics Committee (CAAE: 75982215.2.0000.5054) and followed the Declaration of Helsinki. All participants received an explanation of the study protocol before signing the consent form.

Statistical Analysis

As the total ACE-R score and its subitems and MMSE scores exhibited normal curve distribution in the Kolmogorov Smirnov Test ($p > 0.05$), parametric testing was performed, with Pearson correlation and ANOVA independent group test with Bonferroni correction for multiple comparisons being performed. A p -value < 0.05 was adopted as statistically significant. SPSS version 26.0 was carried out for calculation.

Optimal sensitivity (sn) and specificity (sp) values were defined based on Youden's index⁴³: $J: \max \{sn_i + sp_i - 1\}$, where i represents the pair of coordinates on the graph.

RESULTS

Sociodemographic characteristics

Table 1 shows the main clinical characteristics of the study participants. One hundred participants were evaluated (mean age 73.02; SD 2.89). Mean age was lower among controls

than aMCI and naMCI participants, but this difference did not reach statistical significance. Mean schooling was lower among controls compared to aMCI.

Like our previous study, most participants were married, homemakers, diagnosed with MCI, and had at least two medical comorbidities (hypertension, dyslipidemia, or thyroid disorders). (Table 1). Their mean income was 487.15 US dollars (SD 500.54), an average income level in Brazil⁴⁴.

Correlation between variables

Education was moderately correlated with MMSE ($r=0.491$, $p \leq 0.001$) and ACE-R scores ($r=0.422$, $p \leq 0.001$). In the ACE-R, visuospatial skills were most strongly correlated with educational level ($r = 0.452$, $p < 0.001$). Conversely, all memory subitems, including immediate memory, retrograde, anterograde memory, late recall, and recognition memory, were less influenced by educational level ($p > 0.05$).

Cognitive comparisons among groups

Table 2 depicts cognitive profile and group comparisons. Participants with aMCI were slightly older than controls and naMCI, although the difference was not statistically significant. Controls showed mean MMSE and ACE-R global scores significantly higher than those for naMCI and aMCI participants. Conversely, both MMSE and ACE-R were not significantly different between naMCI and aMCI. Mean ACE-R subitem scores for attention, memory, verbal fluency, language, and visuospatial skills were higher in controls than both aMCI and naMCI participants (Table 2); lower scores in memory component I and memory total scores were found in the aMCI comparatively to the naMCI group. Memory component I comprised immediate recall, retrograde, and anterograde subitems; conversely, the memory component included late recall and recognition (Table 2).

Receiver operating characteristic (ROC) curve analysis for group comparisons

We assessed sn and sp for the MMSE and ACE-R using receiver operating characteristic (ROC) curves (Figure 2 – a, b, and c).

When we compared controls versus naMCI, the area under the ROC curve (AUC) for the MMSE was 0.93, which is considered excellent using Meyers' scale⁴⁵; the MMSE showed 0.95 sn and 0.85 sp (95% CI 0.87-0.98; $p \leq 0.001$) for a cutoff score of 24.5 (Figure 2a). For the **ACE-R**, the AUC was 0.919, which is considered an excellent⁴⁵; it showed 95 sn

and 75 sp (95% CI 0.86-0.97; $p \leq 0.001$) for a cutoff score of 59.5 (Figure 2a).

When we compared controls and aMCI, the AUC for the ACE-R was 0.921, which is considered excellent⁴⁵; it showed 78 sn and 92 sp (95% CI 0.86-0.99; $p \leq 0.001$) for a cutoff score of 53.0 (Figure 2b). For the **MMSE**, the AUC was 0.86, which is considered a good⁴⁵; it showed 0.83 sn and 0.85 sp (95% CI 0.77-0.96; $p \leq 0.001$) for a cutoff score of **24.5** (Figure 2b).

When we compared naMCI versus aMCI, the AUC for the ACE-R was 0.578, which is considered very fair using Meyers' scale⁴⁵; it showed 56 sn and 64 sp (95% CI 0.24-0.61; $p \leq 0.001$) for a cutoff score of 48.5 (Figure 2c). For the MMSE, the AUC was 0.61, which is considered very fair⁴⁵; it showed 0.73

sn and 0.39 sp (95% CI 0.43-0.79; $p \leq 0.001$) for a cutoff score of 22.5 (Figure 2c).

DISCUSSION

Our study assessed the cognitive performance of a sample comprising cognitively healthy, aMCI, and naMCI participants using a comprehensive cognitive tool (ACE-R). The subitem with the strongest correlation with educational level was visuospatial skills. The diagnostic accuracy of the MMSE and ACE-R for aMCI and naMCI were substantially lower than that reported in previous Brazilian studies with

Table 1. Socio demographic characteristics

Variables	Group frequency n (%)			
	Controls	naMCI	aMCI	Total
	60 (60.00)	22 (22.00)	18 (18.00)	100 (100.00)
Gender				
Women	48 (80.00)	19 (86.36)	18 (100.00)	75 (75.00)
Men	12 (20.00)	3 (13.64)	-	15 (15.00)
Ethnicity				
Black African	15 (25.00)	3 (13.64)	5 (27.78)	23 (23.00)
Brown	33 (55.00)	19 (86.36)	13 (72.22)	65 (65.00)
White and Other	12 (20.00)	-	-	12 (12.00)
Marital status				
Married	30 (50.00)	6 (27.27)	7 (38.88)	43 (43.00)
Widowed	16 (26.66)	6 (27.27)	5 (27.77)	27 (27.00)
Divorced	7 (11.66)	4 (18.18)	2 (11.11)	13 (13.00)
Single	7 (11.66)	6 (27.27)	4 (22.22)	17 (17.00)
Previous occupation				
Employed	15 (25.00)	4 (18.18)	4 (22.22)	23 (23.00)
Domestic worker	28 (46.66)	12 (54.54)	11 (61.11)	51 (51.00)
Self employed	15 (25.00)	5 (22.72)	3 (16.67)	23 (23.00)
Employer	2 (3.33)	1 (4.54)	-	3 (3.00)
Comorbidities				
Less than one	11 (16.66)	1 (4.54)	6 (33.33)	17 (17.00)
1 comorbidity	18 (30.00)	5 (22.72)	3 (16.66)	26 (26.00)
≥ 2 comorbidities	22 (36.66)	7 (31.81)	1 (5.55)	30 (30.00)
Hypertension	27 (45.00)	11 (50.00)	2 (11.11)	40 (40.00)
Dyslipidemia	15 (25.00)	2 (9.09)	-	17 (17.00)
Diabetes	12 (20.00)	4 (18.18)	-	16 (16.00)
Thyroid and osteo metabolic diseases	7 (11.66)	4 (18.18)	1 (5.55)	12 (12.00)
Other medical comorbidities	7 (11.66)	3 (13.63)	2 (11.11)	12 (12.00)
Family report of dementia	6 (10.00)	1 (4.55)	1 (5.55)	8 (8.00)
Alcohol use (previous or current use)	8 (13.33)	2 (9.09)	1 (5.55)	11 (11.00)
Tobacco (previous or current use)	5 (8.33)	1 (4.55)	3 (16.67)	9 (9.00)

naMCI – nonamnestic-Mild Cognitive Impairment; aMCI – amnestic-Mild Cognitive Impairment

Table 2. Socio demographic and cognitive profile and group comparisons

Variable	Controls (n = 60)	naMCI (n=22)	aMCI (n = 18)	F	Significance – p level*		
	Mean (SD)	Mean (SD)	Mean (SD)		Con vs. naMCI	Con vs. aMCI	aMCI vs. naMCI
Age	71.33 (7.92)	74.55 (7.96)	76.78 (11.42)	3.183	0.300	0.055	0.696
Education (years)	3.33 (1.43)	2.41 (1.48)	2.00 (1.68)	4.893	0.100	0.018	0.751
MMSE	25.68 (2.09)	20.95 (2.79)	22.06 (2.71)	39.208	<0.001	<0.001	0.315
ACE-R	67.40 (10.49)	50.59 (5.86)	49.28 (7.44)	43.325	<0.001	<0.001	0.894
Attention-orientation	14.70 (1.85)	12.06 (2.05)	12.33 (2.11)	20.299	<0.001	<0.001	1.000
Memory component I	10.23 (2.26)	8.77 (2.49)	6.94 (1.92)	15.485	0.029	<0.001	0.033
Memory component II	6.18 (2.60)	4.73 (1.58)	3.11 (1.41)	14.023	0.028	<0.001	0.064
Memory (total)	16.42 (4.18)	13.50 (3.54)	10.06 (2.77)	20.266	<0.009	<0.001	0.017
Verbal Fluency	7.12 (2.82)	4.32 (2.44)	4.22 (2.16)	13.992	<0.001	<0.001	0.993
Language	18.80 (4.27)	13.27 (2.78)	14.28 (3.61)	20.872	<0.001	<0.001	0.695
Visuospatial skills	10.57 (2.37)	8.18 (2.26)	8.50 (2.38)	11.034	<0.001	0.004	0.905

*p values according to group comparisons after post hoc Bonferroni correction; ACER- Addenbrooke cognitive battery – revised; MMSE – Mini Mental State Exam; naMCI – nonamnestic-Mild Cognitive Impairment; aMCI – amnestic-Mild Cognitive Impairment; Attention-orientation: composite score; Memory component I: immediate recall, retrograde and anterograde; Memory component II: late recall e recognition; Memory total: composite score.

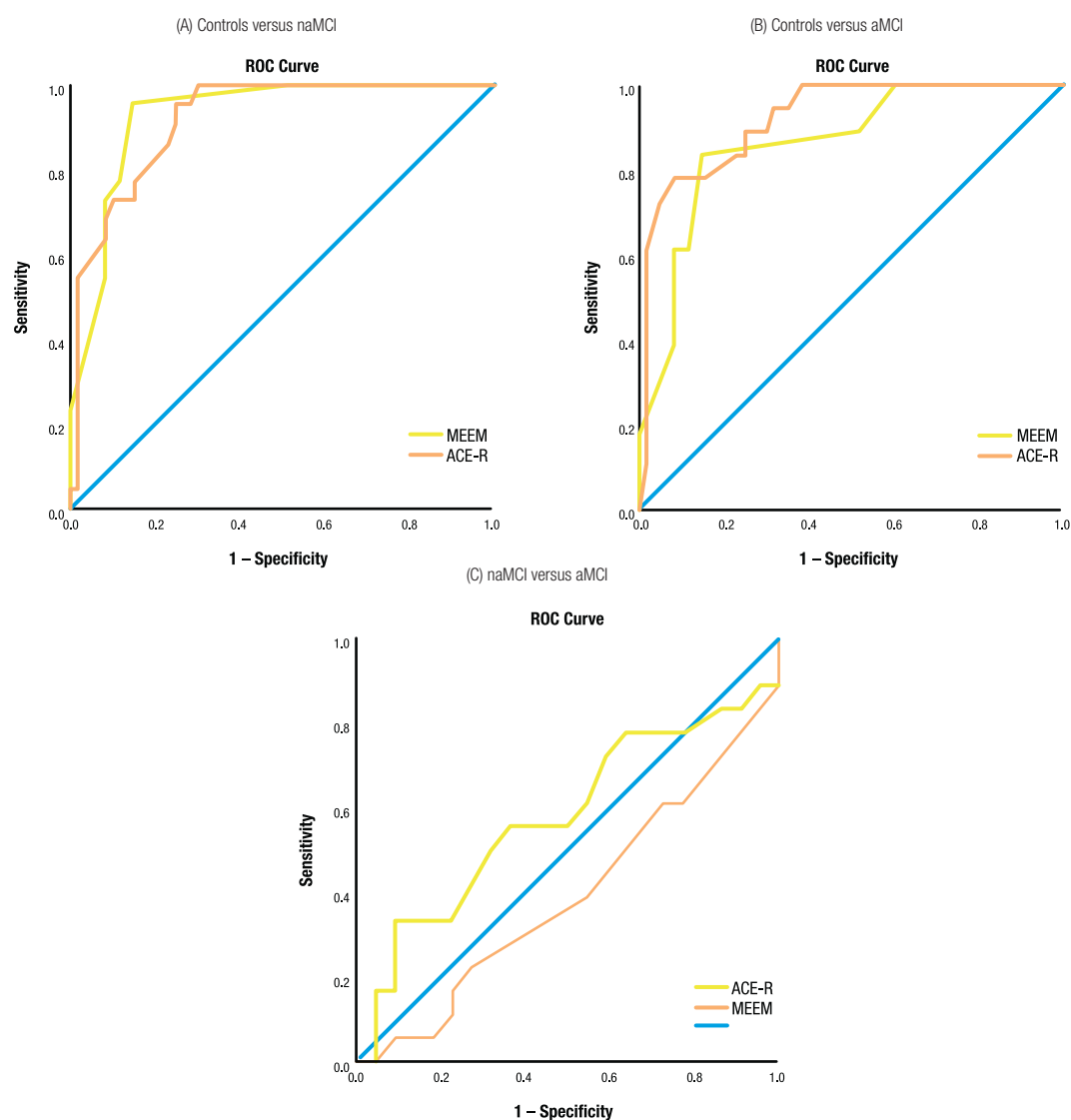
**Figure 2.** ROC curve of MMSE and ACE-R (for statistical details, see results).

Table 3. Accuracy, sensitivity, and specificity of ACE-R

Group Comparison	AUC	Optimal Cutoff	Sensitivity	Specificity	Significance	IC	LR+	LR-
Controls versus naMCI	0.919	<59.5	0.95	0.75	<0.001	0.86-0.97	3.80	0.07
Controls versus aMCI	0.921	<53	0.78	0.92	<0.001	0.86-0.99	9.75	0.24
naMCI versus aMCI	0.578	<48.5	0.56	0.64	<0.001	0.24-0.61	1.56	0.69

naMCI – nonamnesic-Mild Cognitive Impairment; aMCI – amnesic-Mild Cognitive Impairment.

Table 4. Accuracy, sensitivity, and specificity of MMSE

Group Comparison	AUC	Optimal Cutoff	Sensitivity	Specificity	significance	IC	LR+	LR-
Controls versus naMCI	0.927	<24.5	0.95	0.85	<0.001	0.87-0.98	6.33	0.06
Controls versus aMCI	0.863	<24.5	0.83	0.85	<0.001	0.77-0.96	5.53	0.20
naMCI versus aMCI	0.609	≤22.5	0.73	0.39	<0.001	0.43-0.79	1.20	0.69

naMCI – nonamnesic-Mild Cognitive Impairment; aMCI – amnesic-Mild Cognitive Impairment.

ACE-R. Our findings suggest cutoff scores for interpreting ACE-R in a non-dementia sample with low education, considering factors related to the ecological context. Such results can help improve diagnostic accuracy for diagnosing MCI and healthy controls and facilitate early therapeutic interventions.

Overall, mean scores among adults with MCI from both cities in our study (São Luís and Fortaleza) are considerably lower than those described in the literature. International studies, such as Alexopoulos et al., reported higher mean ACE-R scores for cognitive performance³⁰ (controls 90.37 ± 4.99 ; MCI 81.34 ± 9.09). In Brazil, three studies by Carvalho, Caramelli et al. investigated ACE-R performance among Brazilian patients^{26,40,46,47}. In their first study, an adapted ACE-R version was used, and they found a mean total score of 83.3 ± 10.0 for 21 healthy subjects (age 75.4 ± 7.1 ; years of education 8.5 ± 4.3)⁴⁶. In a second study with 144 healthy older adults²⁶, they found higher ACE-R scores among adults 60 to 69 (80.25 ± 9.27) and 70 to 79 (78.75 ± 7.55) when compared to our study (71.33 ± 7.92). Furthermore, the outlined study included more educated participants than our study (mean years of education for the group with 4-7 years of education (4.73 vs. 2.6)²⁶; a comparison of their results with our sample of health controls showed lower scores for all ACE-R domains: memory (17.69 vs. 16.42), verbal fluency (9.42 vs. 7.12), language (22.27 vs. 18.8), visuospatial skills (14.08 vs. 10.57) and attention/orientation (16.79 vs.

14.10)²⁶; in addition, MMSE total scores for healthy controls in our study were also slightly lower (26.52 vs. 25.68)²⁶.

Recently, Carvalho and Caramelli compared the performance between controls and MCI (cutoff = 87, sensitivity = 85%, specificity = 55%), but with a sample of patients > 4 years of education (Table 5)⁴⁷. Another study by Brigola and colleagues⁴⁸ in the municipality of São Carlos, southern Brazil, investigated ACE-R cutoff scores in 667 elderly residents and reported that 67.6% of their sample performed below the literature's suggested values⁴⁸. For individuals between 1-4 years of schooling and 60-69 years of age in the study of Brigola and colleagues⁴⁸, ACE-R performance was higher than our study for ACE-R global scores (60.0 ± 17.7 versus 55.96 ± 19.9) and somehow similar in most ACE-R subdomains: attention orientation (13.4 ± 2.9 versus 13.7 ± 2.3), memory (13.7 ± 5.8 versus 14.6 ± 4.5), verbal fluency (5.4 ± 2.8 versus 5.9 ± 2.9), language (17.6 ± 5.8 versus 16.7 ± 4.6), visuospatial (9.7 ± 3.6 versus 9.7 ± 2.5). However, unlike our results, the study of Brigola and colleagues does not provide further information on the cognitive status of their sample, particularly the possible enrollment of individuals with clinical dementia.

One study in Southeastern Brazil evaluated older adults with different levels of education and included patients with cognitive impairment, no dementia (CIND), AD, and healthy controls. The scores for those with less than five years of education⁴⁰ were: CIND vs. controls: AUC = 0.720

Table 5. Accuracy, sensitivity, and specificity of the ACE-R in the Brazilian literature

Author	Group comparison	AUC	Optimal cutoff	Sensitivity	Specificity	LR+	LR-
Carvalho et al. 2017	Controls versus MCI-AD	0.75	87	1.00	0.46	1.85	0.00
Carvalho et al. 2022	Controls versus MCI	0.71	87	0.85	0.55	1.89	0.27
Tavares-Júnior et al. 2020	Controls versus MCI	0.69	57.5	0.80	0.66	2.35	0.30

MCI – Mild Cognitive Impairment; MCI-AD – MCI who converted to Alzheimer's Dementia over a three-year follow up.

(cutoff <65; sn =76, sp =60). Regarding ACE-R subitems, we cannot compare our data findings with data from this study because they used a different categorization of levels of education in the analysis. Declining scores for illiterate adults or adults with low levels of education are associated with a greater risk of conversion to dementia^{49,50}. Faster cognitive decline has been associated with a higher risk of AD (rate of risk 4.526, 95% confidence interval [95% CI] 2.993, 6.843, $p < 0.001$) and MCI (rate of risk 2.971, 95% CI 1.509, 5.849, $p = 0.002$)⁴⁹. Each added year of education represents a delay in the rate of accelerated decline of around 0.21 years⁵¹; an individual with four years of education may have a rate of accelerated decline before conversion to dementia of around 6.4 years⁵¹.

Our study confirms the importance of stratifying norms for seniors with fewer than four years of education. Inspection of these norms for controls shows that illiterates may perform 40 points less than mates with 12+ years of schooling⁵². Our results also align with previous studies showing that ACE-R remains a viable instrument for lower levels of education as long as education-adjusted values are considered a reference for average performance⁵². In addition, results herein reported for the aMCI group significantly lower performance in memory I components (immediate recall, retrograde, and anterograde) than naMCI. The literature supports these findings, which depict episodic memory as more accurate than recognition memory in the discrimination between later and early presentations of aMCI⁵³ and between aMCI and healthy controls⁵⁴.

Our findings for the ACE-R also reported higher specific values than the literature for the discrimination between controls and both aMCI and naMCI^{40,47}. Clinically, these results suggest the higher specificity of this battery in detecting individuals who do not show the disease⁵⁵. More importantly, our findings point to the need to rationalize the costs of the diagnostic workup of cognitive decline in the community setting, for instance, avoiding ostensive cognitive assessment, biomarkers, or even pharmacological intervention in subjects less likely to decline throughout time^{56,57}.

To the best of our knowledge, this is the first study to assess the performance characteristics of a global cognitive screening tool in MCI subtypes from Northeast Brazil. ACE-R has shown similar sensitivity and accuracy to other comprehensive cognitive instruments, such as CERAD and MoCA, in detecting MCI. It seems particularly important for detecting visuospatial and executive domains¹⁴. Since specific underdiagnosed subtypes of dementia (Lewy bodies and Parkinson's disease dementia) may exhibit significant visuospatial and executive deficits⁵⁸ as well as memory impairment, the use of comprehensive cognitive tests remains desirable, particularly for the tertiary care

setting¹⁴, since visuospatial skills have shown to be strongly associated to lower schooling. Our results offer a more accurate interpretation of visuospatial performance, considering possible educational biases.

Our study has some limitations that deserve consideration. First, we cannot establish cause-effect relationships from cross-sectional data. Second, the concept of MCI also includes multi-domain subtypes not included in our sample study, even though the validity of these and other subdomains of MCI has been questioned by some studies⁵⁹. Third, the participant's level of education was self-reported. Most studies need to consider the quality of education while studying adults with low levels of education, thus preventing an underestimation of the effect of this variable. In addition, more recent research has investigated other variables, including language skills⁵⁰, vocabulary⁶⁰, cognitive reserve⁵¹, abstraction ability, and formal-logical operational capacity deemed more sensitive to establishing educational status.

CONCLUSIONS

Our study assessed ACE-R performance in healthy controls, aMCI, and naMCI subjects. Average scores for healthy aging were considerably lower than those reported in prior Brazilian studies conducted with similar methodology. The analysis of ACE-R diagnostic accuracy between non-dementia groups also evidenced lower cutoff scores compared to benchmark Brazilian studies. Our findings highlight the need for more studies about cognitive changes in Brazil's older adults with low education levels and MCI. The ecological value of these investigations, particularly in geographical areas with poorer economic resources, and potential variables associated with performance, such as cultural characteristics and heterogeneity of illiterate groups, should be considered. These studies can provide additional evidence to support screening approaches and facilitate early diagnosis, optimizing therapeutic intervention for subjects with low schooling and cognitive decline at risk for conversion to dementia.

INDIVIDUAL CONTRIBUTIONS

JSP – sample selection, manuscript writing; **LB**– sample selection, cognitive assessment; **PBN** – study design, manuscript writing, and revision; **JWLTJ** – cognitive assessment; **CA** – cognitive assessment and statistical analysis; **JISN**– study design, manuscript writing, and revision; **FKS** – method design, data discussion; **GSA** – study design, manuscript writing, and final revision

CONFLICT OF INTEREST

No conflicts of interest are to be declared by any of the authors in this study.

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